

In the world of gaming, diffusion models may have a future for generating video game assets to be animated. Although this raises the question of whether these systems could replace human designers, it seems collaboration between humans and AI is the direction both parties are moving in. Graphic designers and digital artists need not panic about advancements in this field.

Furthermore, diffusion models' inpainting capabilities enable automatic photo editing and the iterative creation of complex scenes from scratch, leading to the possible commercialization of such tools. This feature may even make its way into popular photo editing software.

Glossary of important terms

1. **Diffusion model** - a type of generative model that generates a chain of increasingly noisy images by adding Gaussian noise to an image and then training a model to predict the noise added to an image from one step to the following one.
2. **Stable diffusion** - a variant of diffusion models that use a fixed and finite number of steps to generate images, resulting in stable and repeatable image generation.
3. **Residual layers** - a type of layer used in deep neural networks that aim to make the training of deep networks more efficient by allowing gradients to be directly propagated to earlier layers.
4. **Downsampling convolution** - a type of convolutional layer that reduces the spatial resolution of the input feature map.
5. **Upsampling convolution** - a type of convolutional layer that increases the spatial resolution of the input feature map.
6. **Skip connections** - a type of connection in a neural network that allows the output of one layer to be passed directly to a later layer.
7. **Conditioned diffusion** - a technique that involves feeding conditioning data to the diffusion model to aid in producing a sample that corresponds to that data.
8. **Classifier guidance** - a technique that involves adding the gradient with respect to a target class predicted by a classifier to the noise predicted by the diffusion model.
9. **Prompt engineering** - the process of crafting the correct prompts to get a generative model to output exactly what is desired.
10. **Inpainting** - a technique that allows for modification of an image on the command given a text prompt and a mask.